

Would a lot of the players not incline towards hydrogen as a fuel in the near future for this category?

Working with various organisations on hydrogen fuel cell technology, I have observed that importing fuel cells comes at a hefty price—typically five to six times higher than conventional EVs. Despite the allure of a Pune to Mumbai journey via a hydrogen fuel cell bus, the challenges of storing hydrogen pose significant risks, given its explosive nature. In contrast, as lithium prices decrease, it remains the preferred choice in EV and battery storage sectors, overshadowing other alternatives.

But hasn't sodium-ion recently made big news in India?

Over the next decade, lithium will experience the highest rates of adoption in EVs. Sodium faces a significant challenge with high temperatures, resulting in fewer cycles compared to lithium batteries. Additionally, the process of homologating a new technology involves substantial investment of money, effort, and time. This lengthy process often lags behind advancements in already homologated lithium-ion technology.

Can the lithium battery industry grow in India without having a cell manufacturing ecosystem here?

While many battery makers currently import cells from around the world, efforts to establish cell manufacturing capacities are underway in India. Government schemes have been announced, and major corporations such as Reliance and Adani have unveiled their plans in this regard. Although achieving significant yield will require four to five years, this investment is necessary to foster a robust market and ecosystem for EVs in India.

If these are not able to match prices of what they get from say China, why

would battery OEMs buy locally from India?

Today, the taxes and the duties levied on importing these are low. Tomorrow, when India has the capability to manufacture these at par with the quality available from outside, there could be a revision to these taxes and duties. Moreover, the patriotism that is now spreading will definitely encourage Indian companies to buy from other Indian companies.

Why do that, if we can actually make batteries from cells sourced from outside India?

The effectiveness of EV batteries and BMS hinges on their ability to adapt to local climate conditions. Success in one region does not guarantee success elsewhere. Given geopolitical factors, it's imperative for us to develop self-sufficiency in addressing these challenges.

Speaking of self-reliance, would switching to sodium tech mean taking a step towards the same?

Sodium ion technology is now transitioning from the lab to testing and manufacturing stages. While companies like Tesla, Panasonic, Samsung, and LG have dedicated over two decades to lithium technology, only a few have replicated their success. Despite openly available manufacturing recipes, achieving the same level of success remains a challenge.

For instance, many battery OEMs offer warranties of over 6000 cycles, which is impressive. Until we witness a solution successfully enduring 100,000km on the road in temperatures ranging from -5 to 50 degrees Celsius, we cannot claim to have found a viable alternative.

Do you see other bottlenecks in sodium-ion tech?

There might be a ton of supply chain challenges. Good technology does not mean a good supply chain.

For example, hard carbon is a significant challenge hovering over the sodium-ion ecosystem. Reproducibility is also a challenge at the moment. A 100,000km during five years means 5000 cycles, which is also a significant bottleneck.

Is there a difference in the way of designing LFP and sodium-ion packs?

The process remains largely unchanged for battery pack manufacturers. Instead of cells featuring LFP, they will now feature sodium ions. Whether cylindrical or any other form, the manufacturing process remains consistent. Setting up sodium-ion battery pack manufacturing won't require additional investments, but there will be a need for battery cooling systems.

Where do the hybrids fit in these sodium-ion calculations?

Implementing sodium-ion batteries in hybrid vehicles remains a distant possibility at present. Cost is a significant factor in hybrid vehicle setups. For individuals commuting 50km daily, traffic consumes a substantial portion of their vehicle's energy, be it fuel or electric. What is needed in India is the development of compact electric cars that offer excellent value for money.

What about hydrogen as a fuel?

Hydrogen as a fuel holds immense promise for the future. However, its primary challenge lies in storage. Managing a highly volatile liquid under pressures ranging from 300 to 700 PSI poses significant hurdles. Only a few companies worldwide have developed Type 3 and Type 4 cylinders for hydrogen storage, all located outside India. Importing these cylinders would incur substantial costs. It is crucial to remember that environmental conservation initiatives must also offer cost benefits to resonate with people. **EFY**